

# AIMO Proof Pilot — Final Grade

Problem (#\_ID): 5\_POLYGO

Submission: model\_2

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Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

## 1. Final mark

Final mark (choose one): 6

## 2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

An essentially complete solution, taking a slightly different route to the markscheme: it first uses the Rational Root Theorem to show that  $|a_0|$  is the largest coefficient absolute value  $N$ , then via the successive congruences reaches the markscheme's decisive block form, correctly handles the  $N = 1$  case, imposes  $P(1) = 0$ , and counts the admissible odd  $N$  to obtain the correct total of 6150968 (the converse is sufficiently visible from the adjacent cancellations, so **ME5** does not apply).

There is, however, a localised gap in the argument (line 43): the stated reason that  $|\varepsilon_0 + \varepsilon_1 t_1| < t_0$  for every  $t_1 > 1$  is false when  $t_1 = t_0$  and the signs agree. The missing case is easily repaired — if  $t_1 = t_0$  then  $\varepsilon_0 + \varepsilon_1 t_1 \equiv \varepsilon_0 \not\equiv 0 \pmod{t_0}$ , and for a proper odd divisor  $t_1 > 1$  oddness gives  $t_1 \leq t_0/3$  so the intended size argument works — but as written it is a genuine minor proof omission. (The “divide by  $t_0$ ” remark in line 50 is a mere typographical slip, since equation (4) is itself correct.) This is an **ME**-level omission rather than a failure of the main structure, so the otherwise-complete solution scores **6** rather than 7.